

06 MAY 2026

BAYESIAN ANALYSIS

BAYESIAN HIERARCHICAL ANALYSIS

**SOCIOECONOMIC FACTORS
AFFECTING LIFE EXPECTANCY IN
EAST ASIAN AND PACIFIC COUNTRIES**

2000 – 2022 | 17 COUNTRIES | 391 OBSERVATIONS

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Motivation & Research Question

Why EAP Region?

- Rapid economic growth 2000–2022
- Diverse income group (high → lower–middle class)
- Economic growth doesn't explain disparities

Research Focus

- Impact of socioeconomic factors – GDP per capita, health expenditure (%GDP), urbanisation, fertility rate
- Bayesian hierarchical approach
- Linear and Non-linear specifications



Key Variables & Descriptive Statistics

Variables	Mean	Range	Key insight
Life expectancy (years)	72.73	56.36–84.56	Large cross-country spread
Log GDP per capita	8.698	5.69–11.02	Log-transformed (right-skewed)
Health expenditure (% GDP)	4.898	1.77–12.32	Moderate variation
Clean water (%)	88.76	47.01–100.00	High-access ceiling effect
Urban population (%)	54.96	17.50–91.98	Wide urbanisation range
Fertility rate (%)	2.294	0.77–4.56	Strong negative correlate

Dataset

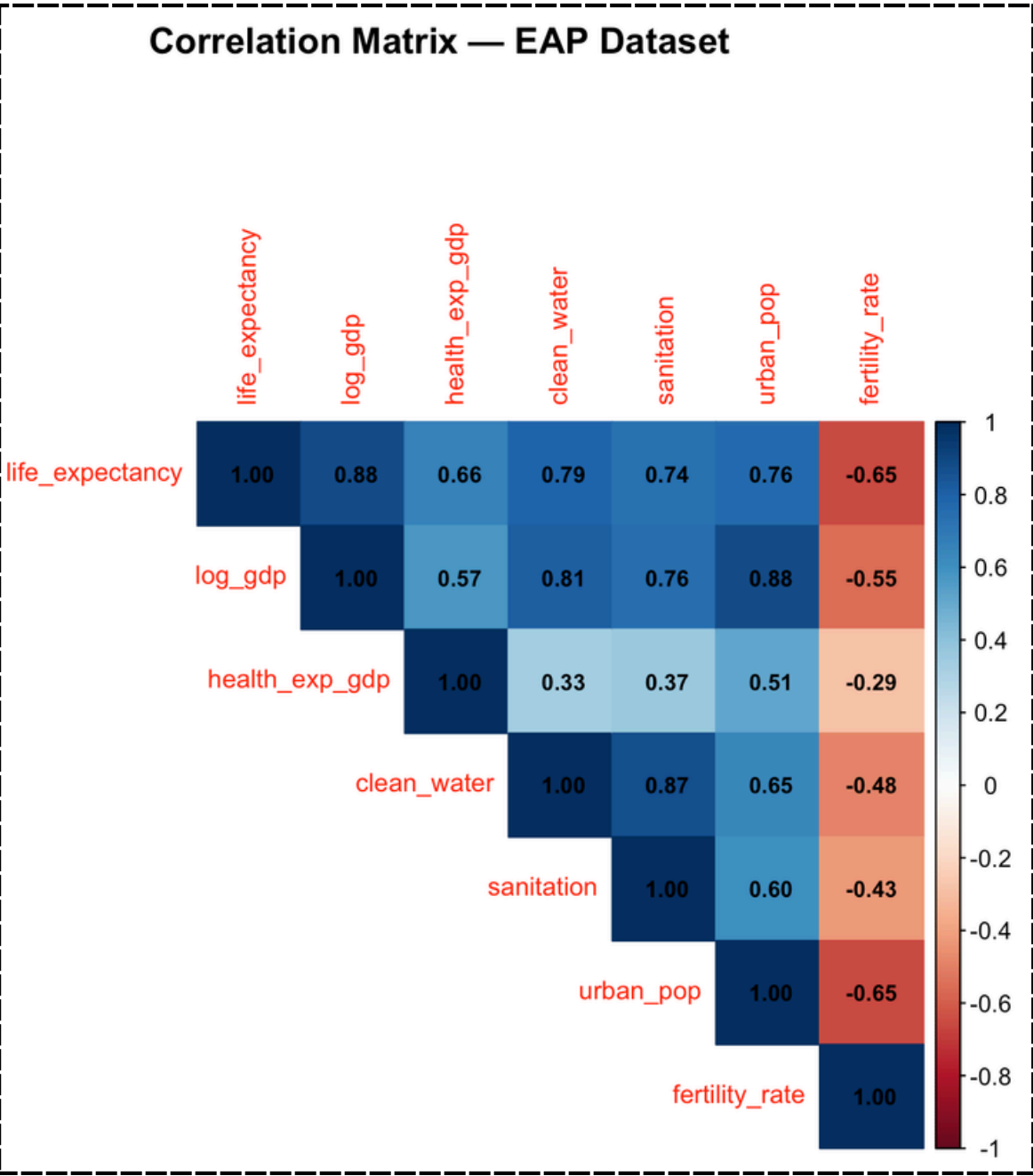
17 EAP Countries (2000–2022)

High Income	Australia, Brunei Darussalam, Japan, New Zealand, South Korea
Upper Middle Income	China, Fiji, Indonesia, Malaysia, Mongolia, Thailand
Lower Middle Income	Cambodia, Lao PDR, Myanmar, Philippines, Samoa, Viet Nam



Exploratory Data Analysis

Key insights



Income Gradient

Median life expectancy:
69 yrs (lower-middle) → 72 yrs (upper-middle) → 81 yrs (high-income).
Variability shrinks at higher income.

Log GDP (r = 0.88)

Strongest predictor. Log transformation normalises heavy right-skew.

Fertility Rate (r = -0.65)

Consistent negative relationship across all income groups. Effect diminishes at very high fertility rates – non-linear pattern confirmed.

Clean Water Access (r = 0.79)

Ceiling effect at high access levels. Quadratic fit clearly outperforms linear – motivates spline specification

Health Expenditure (r = 0.66)

Largely driven by between-group differences; limited within-group predictive power once GDP is controlled.

COVID-19 Disruption

Broad regional dip in life expectancy 2020–2022. Myanmar also shows a sharp drop in 2008 (Cyclone Nargis).



Bayesian Framework	- Parameters treated as random variables. - Full posterior distribution over all plausible values. - Combines prior beliefs with observed data via Bayes' theorem. $P(\theta y) \propto L(y \theta) \cdot \pi(\theta)$ - Direct, intuitive uncertainty quantification.
Hierarchical Structure	17 countries: neither identical nor independent. - Partial pooling: $\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2)$ - Guards against overfitting with only 23 obs/countries. - Captures shared regional context. - Hyper-parameters estimated from data.
Non-linear specification	- Clean water: quadratic fit >> linear - Fertility: steep decline, then levels off - time trend: structural COVID-19 disruption, flucations between some countries. - Penalised cubic regression splines (k=5,6) - LOO-CV confirms $\Delta\text{ELPD} = 150.3$ over linear model

Model

Linear Baseline Model
$y_{it} = \alpha_i + \beta_1 \log_gdp_sit + \beta_2 health_exp_sit + \beta_3 clean_water_sit + \beta_4 fertility_sit + \beta_5 urban_pop_sit + \varepsilon_{it}$ $\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \varepsilon_{it} \sim N(0, \sigma^2)$
Non-linear Model (Final Model)
$y_{it} = \alpha_i + \beta_1 \log_gdp_sit + f_1(clean_water_sit) + f_2(fertility_sit) + f_3(year_sit) + \beta_2 health_exp_sit + \beta_3 urban_pop_sit + \varepsilon_{it}$ $\alpha_i \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \varepsilon_{it} \sim N(0, \sigma^2)$ where: f_1, f_2, f_3 denote penalized cubic regression splines (bs = "cr" in <i>brms</i> syntax).



Prior specification

Parameters	Prior	Justification
Intercept	$N(0,1)$	Weakly informative on standardised scale
Linear coefficient (β)	$N(0,0.5)$	1 SD change $\rightarrow$ at most ~ 0.5 SD shift in outcome
Country-level SD	$Exponential(1)$	Mass near zero; allows larger values if needed
Residual (σ)	$Exponential(1)$	Standard weakly informative
Spline smoothing SD (sds)	$Exponential(1)$	Allows near-linearity if data don't support curvature

Key principle: All predictors standardised ($mean = 0$, $SD = 1$) before fitting, making $N(0, 0.5)$ priors comparable across all covariates. Weakly informative priors regularise without constraining the data from speaking



Model Comparison

LOO-CV Results

Linear Baseline Model

→ REJECTED $\Delta\text{ELPD} = -150.3$

Non-linear Model

→ **SELECTED** $\Delta\text{ELPD} = 0.0$

$\Delta\text{ELPD} = 150.3$
(non-linear)

$\text{SE} = 38.9$
Standard Errors of difference



MCMC Diagnostics

Family: gaussian
Links: mu = identity
Formula: life_exp_s ~ log_gdp_s + s(clean_water_s, bs = "cr", k = 6) + s(fertility_s, bs = "cr", k = 6) + s(year_s, bs = "cr", k = 5) + health_exp_s + urban_pop_s + (1 | country)
Data: df_scaled (Number of observations: 391)
Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
total post-warmup draws = 8000

Smoothing Spline Hyperparameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sds(sclean_water_s_1)	0.11	0.07	0.05	0.28	1.00	3788	4573
sds(sfertility_s_1)	0.12	0.08	0.05	0.31	1.00	3550	5010
sds(syear_s_1)	0.05	0.05	0.01	0.16	1.00	2776	3915

Multilevel Hyperparameters:

~country (Number of levels: 17)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.57	0.11	0.40	0.84	1.00	2169	4025

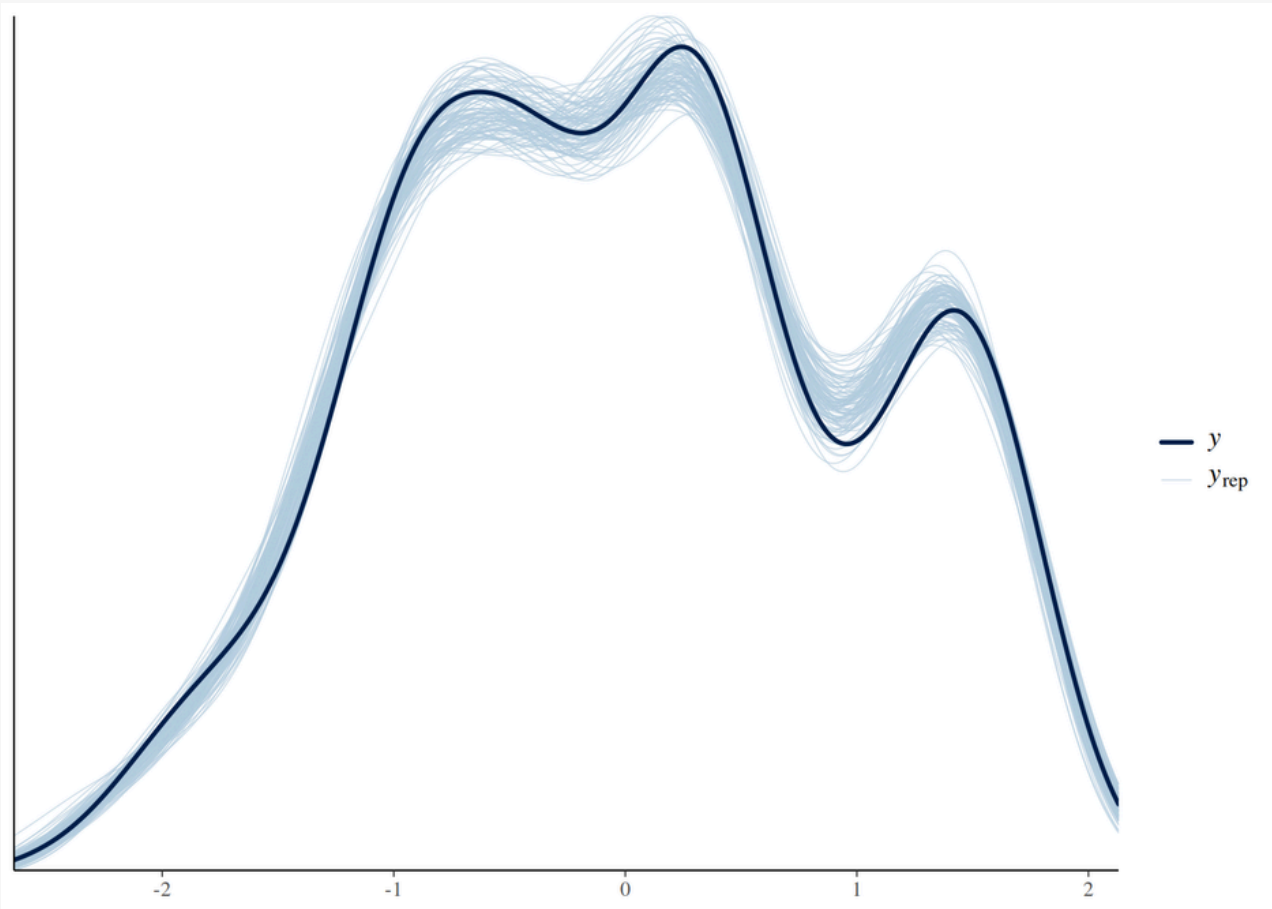
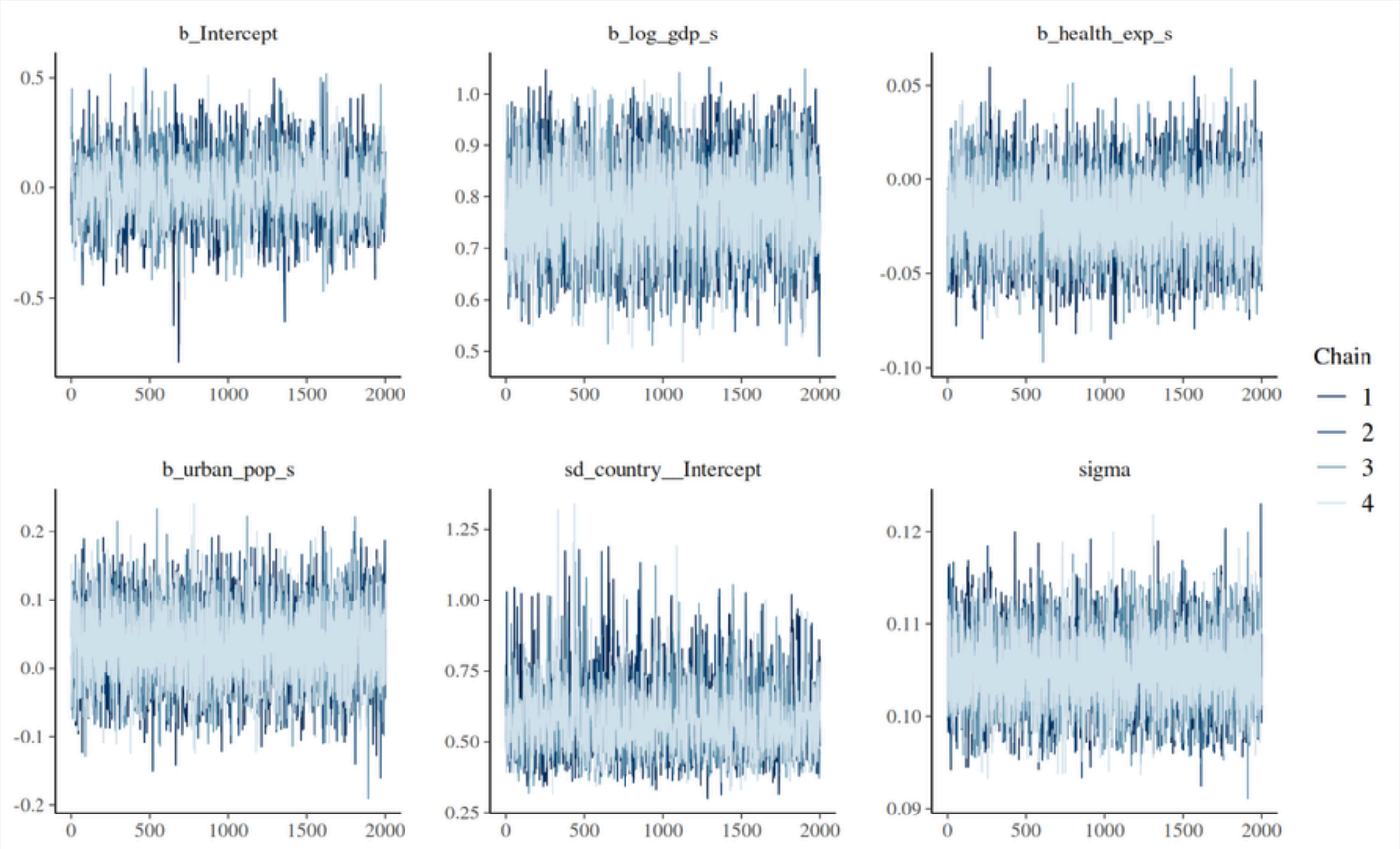
Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.00	0.14	-0.27	0.27	1.00	1936	2716
log_gdp_s	0.78	0.08	0.62	0.94	1.00	5387	5605
health_exp_s	-0.02	0.02	-0.06	0.02	1.00	8598	6358
urban_pop_s	0.04	0.05	-0.07	0.14	1.00	8326	6036
sclean_water_s_1	-0.01	0.01	-0.03	0.01	1.00	6604	6022
sfertility_s_1	-0.09	0.02	-0.14	-0.04	1.00	8488	6277
syear_s_1	0.08	0.01	0.06	0.11	1.00	5055	5405

Further Distributional Parameters:

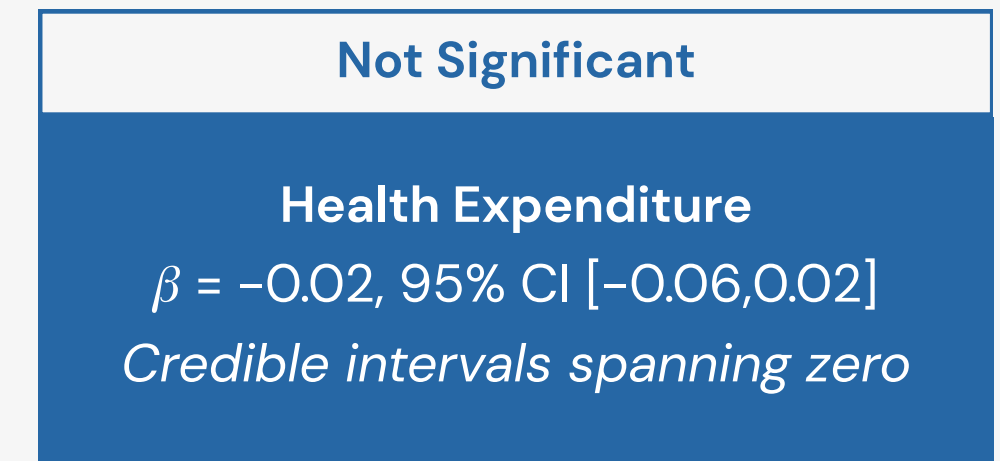
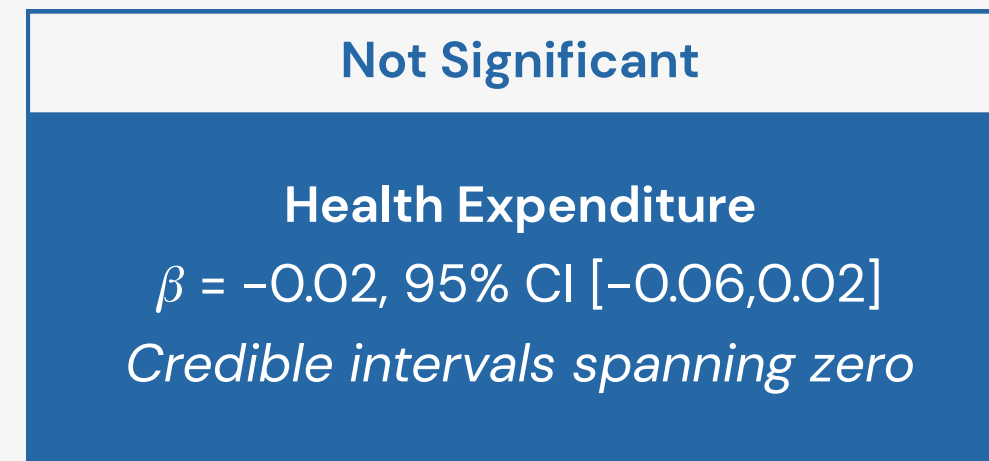
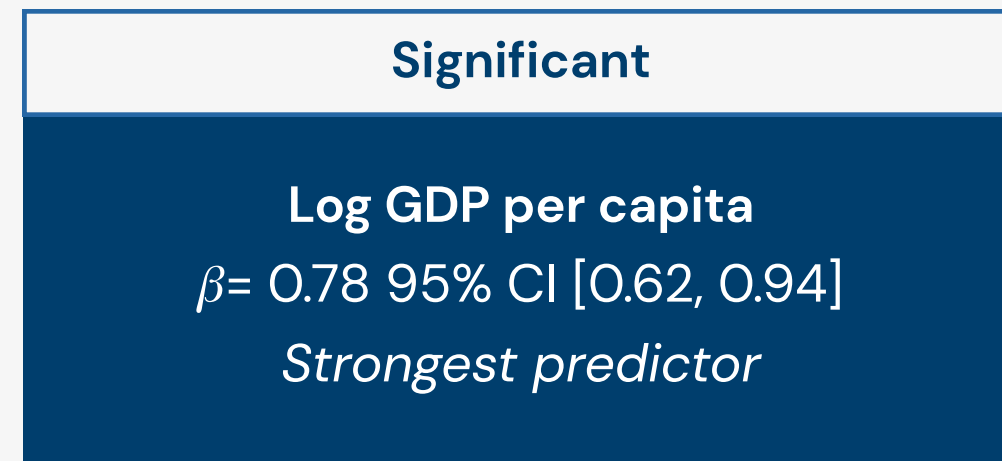
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.11	0.00	0.10	0.11	1.00	10277	6057

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

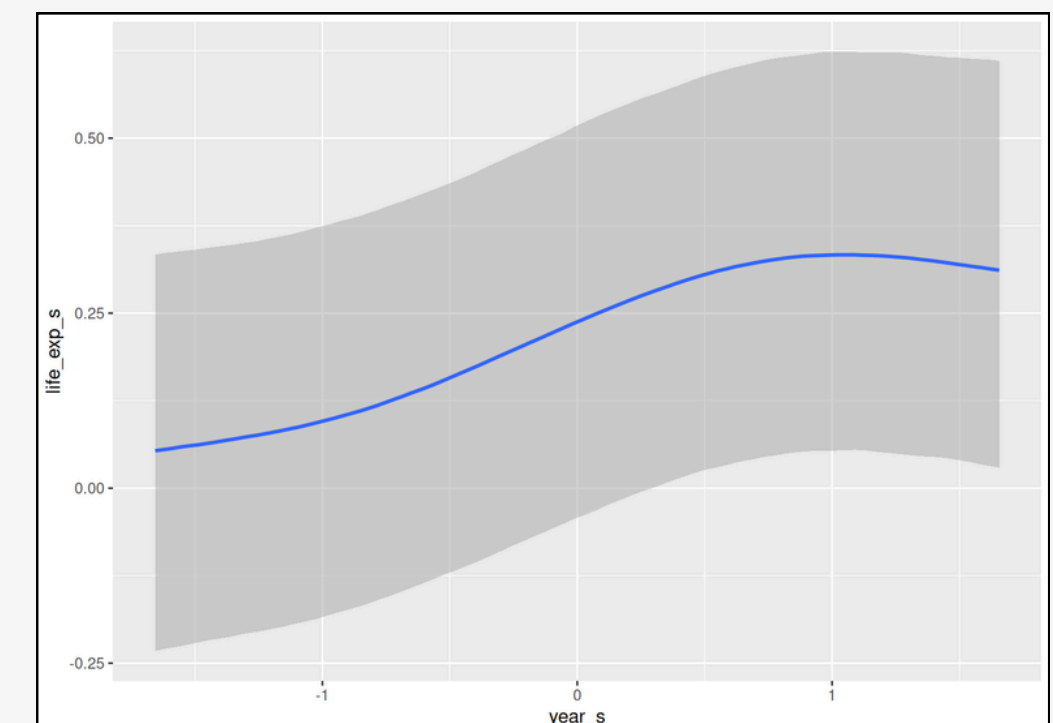
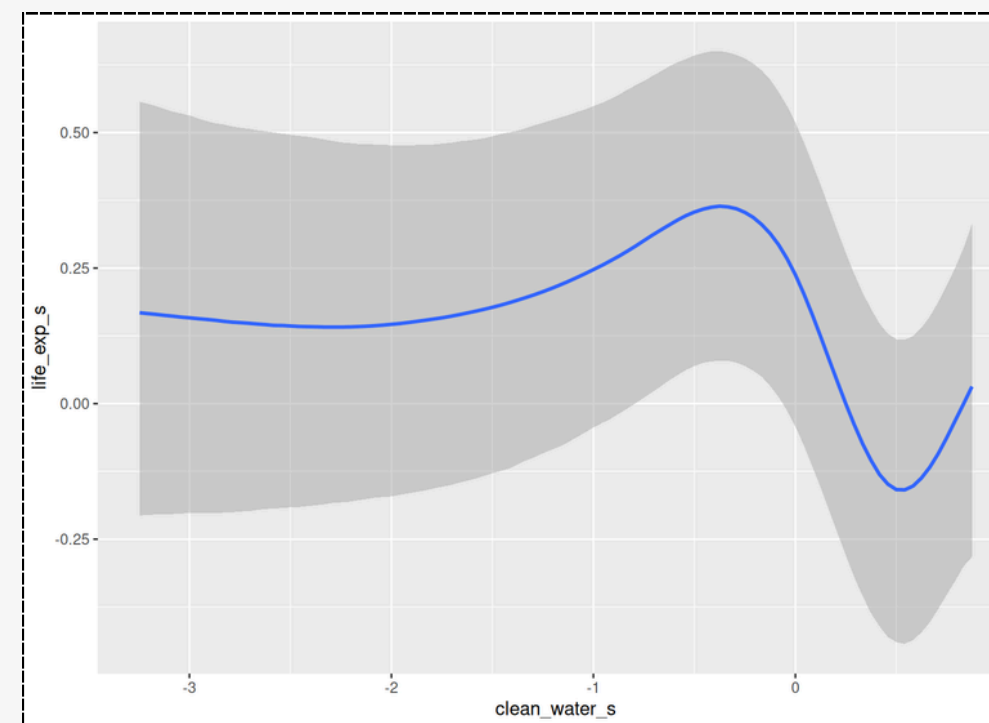
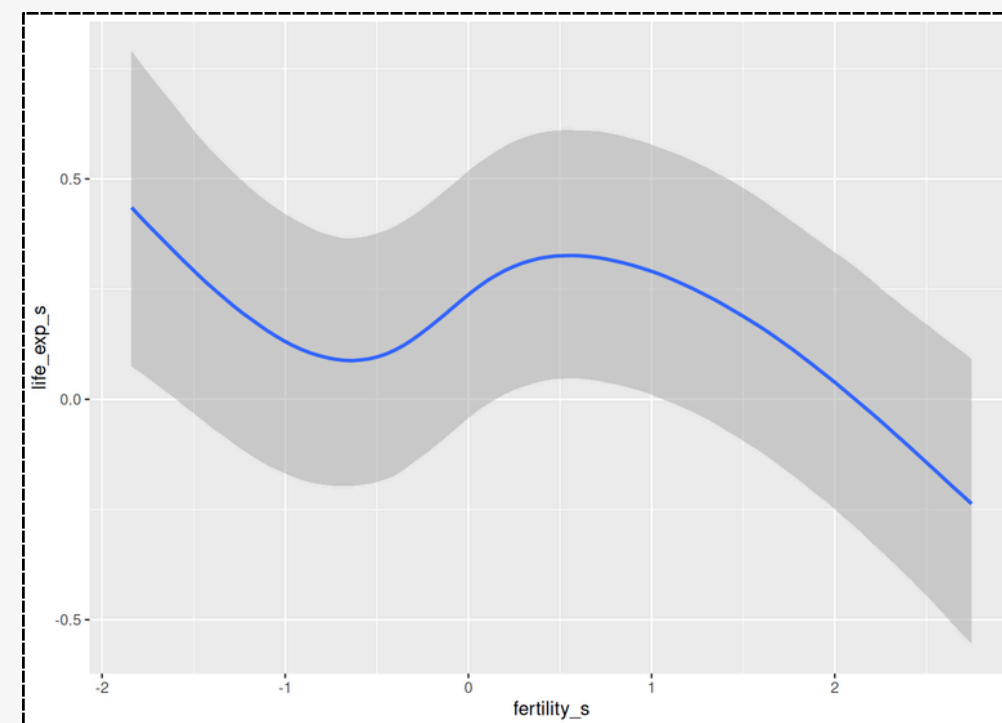


Posterior estimates

Linear Predictors

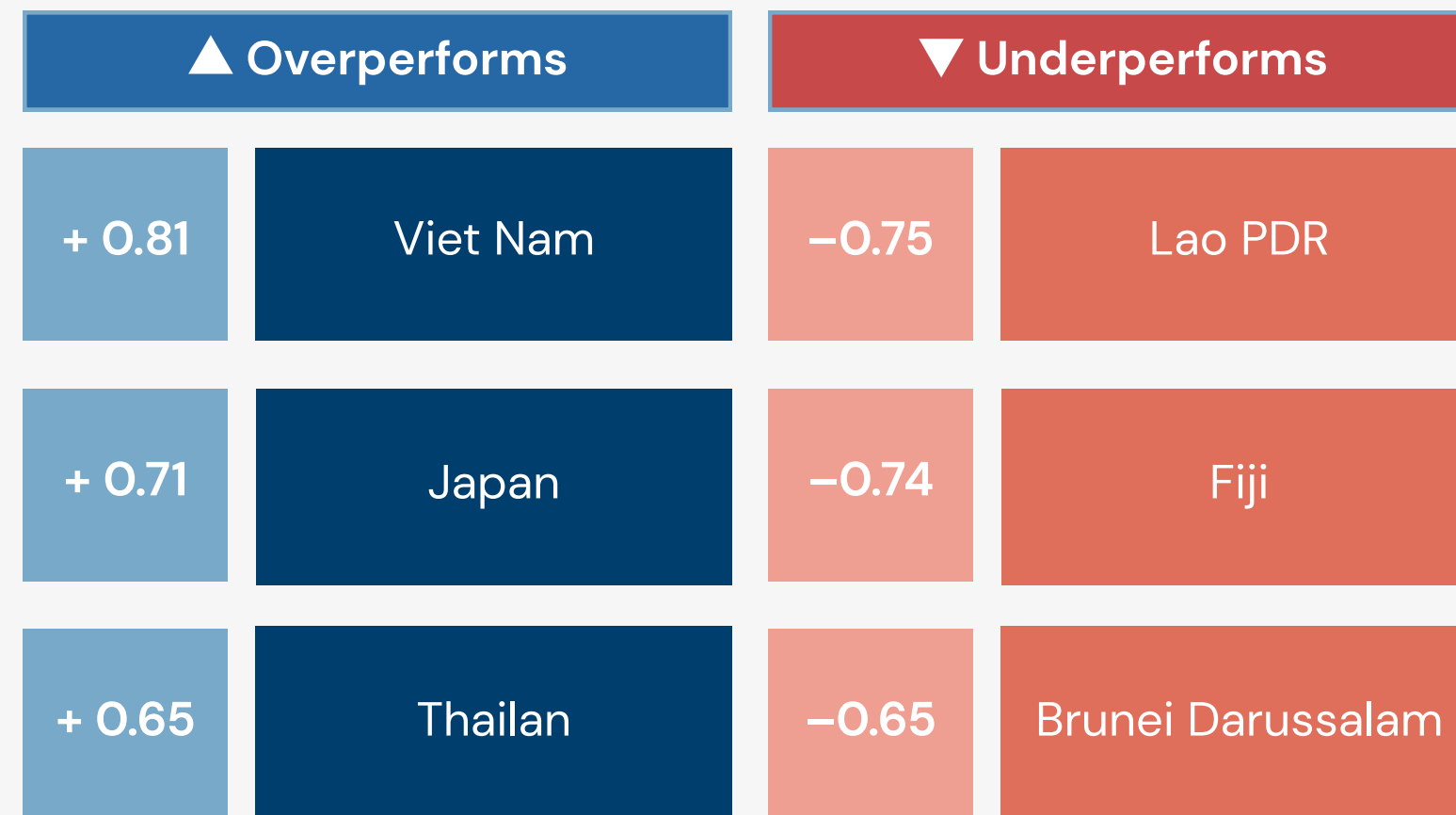


Non-linear Effects: Spline Results



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$country
, , Intercept
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	Estimate	Est.Error	Q2.5	Q97.5
Australia	0.2283655	0.1855626	-0.11996228	0.5984372
Brunei Darussalam	-0.6458314	0.1718400	-0.97550332	-0.3012287
Cambodia	-0.1663586	0.1591368	-0.48601436	0.1399363
China	0.4783037	0.1460173	0.19130802	0.7664656
Fiji	-0.7392155	0.1470394	-1.02743099	-0.4529231
Indonesia	-0.5342858	0.1499619	-0.83166757	-0.2501398
Japan	0.7102675	0.1745533	0.38681346	1.0573153
Korea, Rep.	0.2926546	0.1637625	-0.01731945	0.6223575
Lao PDR	-0.7518577	0.1587252	-1.07506498	-0.4505954
Malaysia	0.2874462	0.1453189	0.00416670	0.5787604
Mongolia	-0.5548276	0.1595625	-0.87130524	-0.2465535
Myanmar	-0.4353304	0.1654771	-0.76888997	-0.1231303
New Zealand	0.2786030	0.1755893	-0.05845929	0.6289343
Philippines	-0.1526038	0.1540915	-0.45511928	0.1424187
Samoa	0.3142621	0.1709910	-0.02524680	0.6444942
Thailand	0.6530938	0.1524836	0.36115555	0.9551753
Viet Nam	0.8073220	0.1528114	0.50986134	1.1036582



Key Findings

Log GDP is Dominant

$\beta = 0.78$ 95% CI [0.62, 0.94] – confirms the Preston curve.

Economic development is the primary structural driver of health outcomes in the EAP region.

Non-linearity is Real

LOO-CV confirms $\Delta\text{ELPD} = 150.3$ (SE 38.9); ~ 3.9 SE above zero.

Penalized splines for fertility, clean water, and time trend outperform linear alternatives.

Substantial Country Heterogeneity

$\text{sd}(\text{Intercept}) = 0.57 \gg \sigma = 0.11$. Unmeasured factors matter.



LIMITATIONS

SAMPLE SIZE Only 17 countries limits generalisability across the broader EAP region.	AGGREGATE DATA Country-level data masks important within-country variation and sub-nation disparities.	OSERVATIONAL ONLY Correlational patterns — causal inference not possible; omitted variable bias cannot be ruled out.
TEMPORAL AUTOCORRELATION Model assumes independent residuals. Within-country residuals likely correlated over time. AR(1) structure warranted.	RANDOM INTERCEPT Slopes constrained to be identical across countries. Random slopes would allow heterogeneous effects by income group.	



BAYESIAN ANALYSIS PROJECT

THANK YOU

FOR YOUR ATTENTION

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